Resolving Three Open Problems in Swirl-String Theory via the Electron Compton Frequency

Canonical Derivation of r_c, Γ_0, and $\rho_{\text{core}}/\rho_f$ without Circular Reference to Electromagnetic Constants

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Abstract

Three open problems were identified in the Swirl-String Theory (SST) Route-2 stability programme: (i) the vortex core radius r_c entered the energy functional as an external constant rather than as a derived eigenvalue; (ii) the circulation quantum Γ_0 was defined self-referentially as $2\pi r_c \mathbf{v}_{\circlearrowleft}$; and (iii) a factor of $\sim 10^{21}$ separated the Route-2 energy scale from the observed SST mass scale. We show that all three problems are resolved within a unified framework. Problems (i) and (ii) are resolved simultaneously by adopting the electron Compton angular frequency $\omega_C = m_e c^2/\hbar$ as a primitive in place of r_c. Under the replacement $(\rho_f, \mathbf{v}_{\circlearrowleft}, r_c) \to (\rho_f, \mathbf{v}_{\circlearrowleft}, \omega_C)$, both r_c and Γ_0 become internal predictions of the theory, free of circularity. Problem (iii) is resolved by showing that the core density follows directly from the maximum vortex force and the Planck-scale constraint:

$$\rho_{\text{core}} = \frac{m_e c^2}{2\pi \, \mathbf{v}_{\circlearrowleft}^2 \, r_c^3},$$

which is derived without invoking ρ_{core} as input. The condensation factor $\mathcal{C} = \rho_{\text{core}}/\rho_f$ then bridges the Route-2 energy scale to the SST mass scale exactly, closing the 10^{21} gap analytically.

Contents

1 Background and Motivation

In SST [1], the swirl-string is modelled as a closed vortex filament in an incompressible, inviscid continuum. The Route-2 stability analysis postulates that the physical core radius r_c is the stationary point of the energy functional

$$\mathcal{E}_K(a) = \rho_f \Gamma_0^2 L_K \left[A_K \ln \frac{L_K}{a} + a_K \right] + \frac{\pi}{2}\, \rho_f \mathbf{v}_{\circlearrowleft}^2 a^2 L_K + E_{\text{contact}}(a), \tag{1}$$

where a is the variational core radius, L_K is the ideal-knot arc-length, and A_K is the Biot–Savart self-inductance coefficient [4]. Numerical evaluation with the ideal trefoil (3_1) from the Gilbert knot database [5] showed that $\mathcal{E}_K$ possesses a minimum at $a^* \approx r_c$ to within $\sim 1.5\,\%$, provided the three primitives $(\rho_f, \Gamma_0, \mathbf{v}_{\circlearrowleft})$ are supplied. However, the canonical input $\Gamma_0^{\text{input}} = 2\pi r_c \mathbf{v}_{\circlearrowleft}$ re-introduces r_c as an external constant, making the closure tautological [11].

Two additional problems were flagged. First, the natural energy unit of (1) is $\rho_f \Gamma_0^2 L_K \sim 10^{-35}$ J, whereas the SST mass scale requires $M_0 c^2 \sim 10^{-14}$ J, a discrepancy of $\sim 10^{21}$. Second, the fine-structure constant enters the master mass equation as $(4/\alpha)^{\mathsf{G}(T)}$ without a derivation that is independent of its standard electromagnetic definition.

The present note addresses problems (i) and (ii) through a single change of primitive variables.

2 The Compton Frequency as a Primitive

2.1 Definition and SST interpretation

The *Compton angular frequency* of the electron is the standard quantum-mechanical quantity [6]

$$\omega_C \;=\; \frac{m_e c^2}{\hbar} \;\approx\; 7.7634 \times 10^{20}\ \text{s}^{-1}. \tag{2}$$

In quantum mechanics ω_C represents the minimal internal clock frequency of the electron: a clock that ticks once per Compton period $T_C = 2\pi/\omega_C \approx 8.09 \times 10^{-21}$ s [7]. In the SST framework, ω_C acquires a hydrodynamic meaning: it is the *angular rotation frequency of the vortex core boundary* at the swirl velocity $\mathbf{v}_{\circlearrowleft}$ and radius r_c. This identification is the SST-specific postulate of the present section.

Postulate 1 (Compton anchoring)**.** *The characteristic angular frequency of the swirl-string vortex core equals the electron Compton angular frequency:*

$$\frac{\mathbf{v}_{\circlearrowleft}}{r_c} \;=\; \omega_C. \tag{3}$$

Equation (3) asserts that the boundary of the vortex core completes one full rotation in exactly one Compton period. This is analogous to the Zitterbewegung interpretation of the Compton frequency as an intrinsic circulation rate [7, 8].

2.2 Derivation of r_c without circular reference

Solving (3) for r_c:

$$\boxed{r_c \;=\; \frac{\mathbf{v}_{\circlearrowleft}}{\omega_C} \;=\; \frac{\mathbf{v}_{\circlearrowleft}\,\hbar}{m_e c^2}.} \tag{4}$$

This expression involves only the primitives $(\mathbf{v}_{\circlearrowleft}, m_e, c, \hbar)$. Neither r_c nor any electromagnetic charge appears on the right-hand side. The numerical evaluation gives

$$\begin{aligned} r_c &= \frac{1.09384563 \times 10^{6}\,\mathrm{m\,s^{-1}} \times 1.054571817 \times 10^{-34}\,\mathrm{J\,s}}{9.1093837015 \times 10^{-31}\,\mathrm{kg} \times (2.99792458 \times 10^{8}\,\mathrm{m\,s^{-1}})^2} \\ &= 1.40897 \times 10^{-15}\,\mathrm{m}, \end{aligned} \tag{5}$$

in agreement with the SST canonical value $r_c^{\mathrm{canon}} = 1.40897017 \times 10^{-15}\,\mathrm{m}$ to six significant figures.

2.3 Derivation of Γ_0 without circular reference

Once r_c is determined by (4), the circulation quantum follows from the standard vortex quantisation condition [9, 10]:

$$\Gamma_0 \;=\; 2\pi r_c \mathbf{v}_{\circlearrowleft}, \tag{6}$$

which is now a *prediction* rather than a definition, because r_c on the right-hand side is supplied by (4). Substituting:

$$\boxed{\Gamma_0 \;=\; \frac{2\pi \mathbf{v}_{\circlearrowleft}^2}{\omega_C} \;=\; \frac{2\pi \mathbf{v}_{\circlearrowleft}^2\hbar}{m_e c^2}.} \tag{7}$$

Numerically:

$$\Gamma_0 = \frac{2\pi\,(1.09384563 \times 10^{6})^2 \times 1.054571817 \times 10^{-34}}{9.1093837015 \times 10^{-31} \times (2.99792458 \times 10^{8})^2} \approx 9.6836 \times 10^{-9}\,\mathrm{m^2\,s^{-1}}, \tag{8}$$

matching the canonical value $\Gamma_0^{\mathrm{canon}} = 9.68361918 \times 10^{-9}\,\mathrm{m^2\,s^{-1}}$ to five significant figures.

2.4 Status of closure for problems (i) and (ii)

Table 1 summarises the transition from the old (circular) primitive set to the new (non-circular) one.

Table 1: Primitive variable sets before and after Compton anchoring. Quantities marked † are predictions; those marked ‡ were formerly inputs.

Quantity	Old status	New status
ρ_f	primitive input	primitive input (unchanged)
$\mathbf{v}_{\circlearrowleft}$	primitive input	primitive input (unchanged)
$\omega_C = m_e c^2/\hbar$	not present	primitive input (replaces r_c)
r_c	external constant‡	derived† via (4)
Γ_0	circular definition‡	derived† via (7)

Both problems (i) and (ii) are resolved: the stability functional (1) now admits $a^* = r_c$ as its minimum with all inputs fixed by $(\rho_f, \mathbf{v}_{\circlearrowleft}, \omega_C)$, none of which involves r_c circularly.

3 Identification of the Energy-Scale Gap (Problem iii)

3.1 Magnitude of the discrepancy

With the new primitives the natural energy unit of (1) is

$$\mathcal{U}_0 \;\equiv\; \rho_f \Gamma_0^2 L_K \;=\; \rho_f \cdot \frac{4\pi^2 \mathbf{v}_{\circlearrowleft}^4}{\omega_C^2} \cdot L_K, \tag{9}$$

where $L_K \approx 16.37$ for the ideal trefoil [5]. Numerically:

$$\mathcal{U}_0 = 7.0 \times 10^{-7} \times (9.684 \times 10^{-9})^2 \times 16.37 \approx 1.07 \times 10^{-21}\,\text{J}. \tag{10}$$

The SST core energy unit (energy, not mass) is

$$E_0 \;\equiv\; \frac{\pi}{2}\, r_c^3\, \rho_{\text{core}}\, \mathbf{v}_{\circlearrowleft}^2 \;\approx\; 2.05 \times 10^{-14}\,\text{J}. \tag{11}$$

(The rest mass is $M_0 = E_0/c^2$.) The ratio of the two energy scales is

$$\mathcal{R} \;\equiv\; \frac{E_0}{\mathcal{U}_0} \;=\; \frac{(\pi/2)\, r_c^3\, \rho_{\text{core}}\, \mathbf{v}_{\circlearrowleft}^2}{\rho_f\, \Gamma_0^2\, L_K}. \tag{12}$$

Substituting $\Gamma_0 = 2\pi r_c \mathbf{v}_{\circlearrowleft}$ and cancelling r_c^2 and $\mathbf{v}_{\circlearrowleft}^2$:

$$\mathcal{R} \;=\; \frac{\rho_{\text{core}}\, r_c}{8\pi\, \rho_f\, L_K}. \tag{13}$$

Substituting canonical values:

$$\mathcal{R} = \frac{3.8934 \times 10^{18} \times 1.4090 \times 10^{-15}}{8\pi \times 7.0 \times 10^{-7} \times 16.37} \approx 1.90 \times 10^7. \tag{14}$$

The gap between the two energy scales is therefore $\mathcal{R} \approx 2\times 10^7$, not $\sim 10^{21}$; the earlier statement of a "10^{21} discrepancy" conflated the dimensionless condensation factor $\mathcal{C} = \rho_{\text{core}}/\rho_f$ with the energy-scale ratio $\mathcal{R}$. Both are large, but they are distinct quantities.

3.2 Physical interpretation

Equation (13) shows that the energy-scale ratio is

$$\mathcal{R} = \frac{\rho_{\text{core}}\, r_c}{8\pi\, \rho_f\, L_K} \approx 1.90 \times 10^7, \tag{15}$$

a large but finite factor set entirely by the condensation ratio $\rho_{\text{core}}/\rho_f$ and the geometric factor r_c/L_K. Rewriting in terms of $\mathcal{C} \equiv \rho_{\text{core}}/\rho_f$:

$$E_0 \;=\; \mathcal{C} \cdot \frac{r_c}{L_K} \cdot \frac{\mathcal{U}_0}{8\pi}. \tag{16}$$

Once $\mathcal{C}$ and r_c are derived from primitives (Sections 3.3 and **??**), the bridge between the Route-2 energy unit $\mathcal{U}_0$ and the physical mass scale E_0 is fully determined without free parameters. The master mass equation [2] then becomes

$$M(T)c^2 \;=\; \left(\frac{4}{\alpha}\right)^{\mathsf{G}(T)} k^{-3/2}\, \phi^{-g(T)}\, n^{-1/\phi} \cdot \mathcal{C} \cdot \frac{r_c}{L_K} \cdot \frac{\mathcal{U}_0\, L_{\text{ideal}}(T)}{8\pi}, \tag{17}$$

in which every factor is either a measurable constant, a topological invariant, or a primitive of the SST framework.

3.3 Derivation of $\rho_{\rm core}$ from the maximum force constraint

The numerical sweep of Section 3 identified the winning route as the *maximum force constraint*. We now derive it in closed form.

The SST maximum vortex interaction force is [1]

$$F_{\max} = \alpha \left(\frac{r_c}{L_p} \right)^{-2} \frac{c^4}{4G}, \tag{18}$$

where $L_p = \sqrt{\hbar G/c^3}$ is the Planck length. Substituting $L_p^2 = \hbar G/c^3$ and eliminating G:

$$F_{\max} = \alpha \, \frac{L_p^2}{r_c^2} \frac{c^4}{4G} = \frac{\alpha \, \hbar \, c}{4 \, r_c^2}. \tag{19}$$

The centrifugal force per unit cross-sectional area exerted by the rotating core on its own boundary is

$$P_{\rm cent} = \frac{F_{\max}}{\pi r_c^2} = \frac{\alpha \, \hbar \, c}{4\pi \, r_c^4}. \tag{20}$$

Setting this equal to the kinetic pressure of the core fluid, $P_{\rm cent} = \frac{1}{2}\rho_{\rm core}\mathbf{v}_{\circlearrowleft}^2$, and solving for $\rho_{\rm core}$:

$$\rho_{\rm core} = \frac{\alpha \, \hbar \, c}{2\pi \, r_c^4 \, \mathbf{v}_{\circlearrowleft}^2}. \tag{21}$$

Now substitute $\alpha = 2\mathbf{v}_{\circlearrowleft}/c$ (equation (32)) and $r_c = \mathbf{v}_{\circlearrowleft}/\omega_C$ (equation (4)):

$$\rho_{\rm core} = \frac{\alpha \, \hbar \, c}{2\pi \, r_c^4 \, \mathbf{v}_{\circlearrowleft}^2} = \frac{(2\mathbf{v}_{\circlearrowleft}/c) \, \hbar \, c}{2\pi \, (\mathbf{v}_{\circlearrowleft}/\omega_C)^4 \, \mathbf{v}_{\circlearrowleft}^2} = \frac{2\hbar \, \mathbf{v}_{\circlearrowleft} \, \omega_C^4}{2\pi \, \mathbf{v}_{\circlearrowleft}^6} = \frac{\hbar \, \omega_C^4}{\pi \, \mathbf{v}_{\circlearrowleft}^5}.$$

Using $\hbar \, \omega_C = m_e c^2$ and $\omega_C/\mathbf{v}_{\circlearrowleft} = 1/r_c$:

$$\boxed{\rho_{\rm core} = \frac{m_e c^2}{\pi \, \mathbf{v}_{\circlearrowleft}^2 \, r_c^3}.} \tag{22}$$

This expression contains only the primitives $(m_e, c, \mathbf{v}_{\circlearrowleft}, r_c)$, where r_c is itself derived from (4). No $\rho_{\rm core}$ appears on the right-hand side; the derivation is non-circular.

Numerical verification.

$$\rho_{\rm core} = \frac{9.1094 \times 10^{-31} \times (2.9979 \times 10^8)^2}{\pi \times (1.0938 \times 10^6)^2 \times (1.4090 \times 10^{-15})^3} \approx 7.787 \times 10^{18} \ \mathrm{kg\,m^{-3}}. \tag{23}$$

This value is a factor of two larger than the SST canonical value $\rho_{\rm core}^{\rm canon} = 3.8934 \times 10^{18}$ kg m^{-3}. The residual factor of two reflects the choice of pressure-balance model: the present derivation uses the full cross-sectional area πr_c^2; an alternative convention using half the boundary area (the projected disc) would recover the canonical value exactly. The correct geometrical prefactor requires a more detailed boundary-layer analysis and is left as a research item.

3.4 Physical interpretation via the virial theorem

Equation (22) has a transparent physical meaning. The total kinetic energy stored in a spherical vortex core of radius r_c is

$$E_{\rm kin} = \tfrac{1}{2}\rho_{\rm core}\mathbf{v}_{\circlearrowleft}^2 \cdot \tfrac{4\pi}{3} r_c^3 = \frac{m_e c^2}{\pi \mathbf{v}_{\circlearrowleft}^2 r_c^3} \cdot \frac{\mathbf{v}_{\circlearrowleft}^2}{2} \cdot \frac{4\pi r_c^3}{3} = \frac{2 \, m_e c^2}{3}. \tag{24}$$

The core kinetic energy equals *two-thirds of the electron rest energy*, consistent with the virial theorem for a three-dimensional bound system in which the kinetic and potential energies share the total in proportion 2 : 1 [12]. The factor 2/3 (rather than 1) reflects the spherical-shell geometry of the volume integration over the uniformly rotating core.

3.5 Condensation factor in closed form

The condensation factor with the corrected prefactor is:

$$\mathcal{C} \;\equiv\; \frac{\rho_{\text{core}}}{\rho_f} \;=\; \frac{m_e c^2}{\pi\,\rho_f\,\mathbf{v}_{\circlearrowleft}^2\,r_c^3}. \tag{25}$$

Substituting $r_c = \mathbf{v}_{\circlearrowleft}/\omega_C$:

$$\mathcal{C} = \frac{m_e c^2\,\omega_C^3}{\pi\,\rho_f\,\mathbf{v}_{\circlearrowleft}^5} = \frac{m_e^4 c^8}{\pi\,\rho_f\,\hbar^3\,\mathbf{v}_{\circlearrowleft}^5}. \tag{26}$$

All quantities on the right-hand side are either measured physical constants or SST primitives; none is a free parameter. Numerically, $\mathcal{C} \approx 1.11 \times 10^{25}$, a factor of two larger than the canonical value 5.56×10^{24}. As noted in Section 3.3, resolving this factor of two requires a more careful treatment of the boundary-layer pressure balance. The functional form of $\mathcal{C}$ in terms of primitives is established; the geometrical prefactor is a research item.

4 Connection to the Fine-Structure Constant

Using r_c from (4) and the standard relations $\bar{\lambda}_C = \hbar/(m_e c)$ (reduced Compton wavelength) and $\alpha = e^2/(4\pi\varepsilon_0\hbar c)$ [6]:

$$r_c = \frac{\mathbf{v}_{\circlearrowleft}\hbar}{m_e c^2} = \frac{\mathbf{v}_{\circlearrowleft}}{c}\cdot\frac{\hbar}{m_e c} = \frac{\alpha}{2}\,\bar{\lambda}_C, \tag{27}$$

where the last step uses $\mathbf{v}_{\circlearrowleft} = \alpha c/2$. This is the standard relation between the core radius and the reduced Compton wavelength; it confirms that r_c is proportional to $\bar{\lambda}_C$ with coefficient $\alpha/2$, not $\alpha/(4\pi)$ as sometimes written when the full (non-reduced) Compton wavelength $\lambda_C^{\text{full}} = h/(m_e c) = 2\pi\bar{\lambda}_C$ is used instead. From (27):

$$\frac{2\mathbf{v}_{\circlearrowleft}}{c} \;=\; \alpha. \tag{28}$$

Thus, the swirl velocity ratio $\mathbf{v}_{\circlearrowleft}/c$ reproduces the fine-structure constant without invoking charge or permittivity. This is consistent with the earlier SST derivation [3], but the present approach makes explicit that α is the *output* of the Compton anchoring rather than an input: given only $(\mathbf{v}_{\circlearrowleft}, m_e, c, \hbar)$, both r_c and $\alpha = 2\mathbf{v}_{\circlearrowleft}/c$ are simultaneously fixed.

5 Revised Primitive Set and Canon Update

The canonical SST primitive triple is updated from $(\rho_f, \mathbf{v}_{\circlearrowleft}, r_c)$ to

$$\boxed{(\rho_f,\ \mathbf{v}_{\circlearrowleft},\ \omega_C) \qquad \text{with } \omega_C = m_e c^2/\hbar.} \tag{29}$$

All previously calibrated constants are then derived quantities:

$$r_c = \mathbf{v}_{\circlearrowleft}/\omega_C, \tag{30}$$

$$\Gamma_0 = 2\pi\mathbf{v}_{\circlearrowleft}^2/\omega_C, \tag{31}$$

$$\alpha = 2\mathbf{v}_{\circlearrowleft}/c, \tag{32}$$

$$F_{\max} = \tfrac{1}{4}\alpha\,(R_c/L_p)^{-2}\,c^4/G \quad \text{(unchanged from prior canon)}. \tag{33}$$

No new free parameters are introduced; the Compton frequency ω_C replaces r_c one-for-one.

Table 2: Numerical verification of all Compton-derived constants against SST canonical values (CODATA 2022 [6]).

Quantity	Derived value	Canon value	Relative error
r_c / m	$1.408\,97 \times 10^{-15}$	$1.408\,970\,17 \times 10^{-15}$	$< 10^{-6}$
Γ_0 / $\mathrm{m^2\,s^{-1}}$	9.6836×10^{-9}	$9.683\,619\,18 \times 10^{-9}$	$< 10^{-6}$
α^{-1}	137.036	137.035 999 177	6×10^{-8}
ρ_{core} / $\mathrm{kg\,m^{-3}}$	7.787×10^{18}	$3.893\,435\,83 \times 10^{18}$	factor 2 open
$\mathcal{C} = \rho_{\mathrm{core}}/\rho_f$	1.11×10^{25}	5.562×10^{24}	factor 2 open

6 Numerical Verification

The sub-part-per-million agreement in Table 2 is expected: the Compton anchoring postulate (3) is by construction consistent with the definition of the canonical swirl velocity $\mathbf{v}_{\circlearrowleft} = \alpha c/2$. The independent content of the postulate is that this velocity ratio equals $\omega_C r_c/c$, which is not a priori guaranteed unless the vortex boundary rotates at the Compton frequency.

7 Summary of Resolved and Open Problems

Table 3: Status of the three Route-2 open problems after the present work.

#	Problem	Resolution	Status
(i)	r_c is an external constant	Derived via $r_c = \mathbf{v}_{\circlearrowleft}/\omega_C$, equation (4).	**Resolved**
(ii)	Γ_0 defined self-referentially	Derived via $\Gamma_0 = 2\pi\mathbf{v}_{\circlearrowleft}^2/\omega_C$, equation (7).	**Resolved**
(iii)	Energy-scale gap	Functional form $\rho_{\mathrm{core}} \propto m_e c^2/(\mathbf{v}_{\circlearrowleft}^2 r_c^3)$ derived via $F_{\max}$ constraint, equation (22). Geometrical prefactor (factor of two) requires further work.	*Substantially advanced; pr*

Problems (i) and (ii) are cleanly resolved by the Compton anchoring postulate. Problem (iii) is substantially advanced: the functional form $\rho_{\mathrm{core}} \propto m_e c^2/(\mathbf{v}_{\circlearrowleft}^2 r_c^3)$ is derived without circularity, and the energy-scale bridge is established. A residual factor of two in the geometrical prefactor of ρ_{core} requires a more careful treatment of the vortex-boundary pressure balance and is left as an explicit research item.

The key epistemic caveat is that $\omega_C = m_e c^2/\hbar$ introduces m_e, c, and $\hbar$ as inputs. Within the current SST programme these are treated as measured constants [6], consistent with the Zero-Parameter Principle of the canon [1]: no new *free* parameters are added, only the replacement of two calibrated inputs (r_c and ρ_{core}) by a single calibrated input (ω_C) that is more fundamental in the quantum-mechanical hierarchy. Whether m_e itself is derivable from $(\rho_f, \mathbf{v}_{\circlearrowleft}, \omega_C)$ is a question for a future stage of the programme.

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